

Project Definition Document: CS5998 Capstone Project

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Program: Master of Data Science & Artificial Intelligence

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1. Problem Statement

Business Context: Customer churn is a critical challenge for telecommunications companies. Retaining an existing customer costs 5-7 times less than acquiring a new one.

Problem Definition: The project addresses the **binary classification** problem: predicting whether a customer will churn in the near future, based on their demographic, service, and account information.

Key Benefits: Proactive intervention, efficient resource optimization, and business intelligence on attrition drivers.

Scope Definition (Bounded Choices):

- **Problem Type:** Binary Classification (Yes/No outcome).
- **Data Type:** Tabular (structured customer records).
- **Technique Category:** Machine Learning.
- **System Context:** Prediction Pipeline + Interpretation Dashboard.

2. Data Source(s)

Source: IBM Sample Data Sets - Telco Customer Churn, available on Kaggle.

Overview:

- **Total Records:** 7,043 customers.
- **Features:** 21 columns covering Demographics, Services, and Account information.
- **Class Distribution:** Imbalanced (~73.5% No Churn, ~26.5% Churn).

Data Quality: 99.8% complete (11 missing TotalCharges values) and appropriate for classification.

3. Intended Methods (Linked to Program Modules)

The project follows a structured machine learning pipeline:

Phase	Methods	Related Module(s)
Data Preparation	Descriptive statistics, handling missing values, encoding, and scaling.	Data Analytics, Data Engineering, ML Fundamentals.
Model Development	Baseline Model: Logistic Regression. Improved Models: Random Forest and XGBoost with hyperparameter tuning.	Machine Learning, Advanced ML.
Imbalance Handling	Techniques such as SMOTE, class weights, and stratified sampling.	Machine Learning.
Interpretation	Explainable AI methods, specifically SHAP values, and feature importance analysis.	Explainable AI.

4. Expected Outputs

Technical Deliverables:

- Preprocessed Dataset.
- Trained Models (baseline and improved).
- End-to-end Prediction Pipeline.
- Interpretation Analysis (SHAP plots).

Documentation Deliverables:

- EDA and Model Development Notebooks.
- Final Report (25-30 pages).
- Demo Video (6-minute walkthrough).

Evaluation Plan:

Given the class imbalance, the validation approach uses stratified train-test splits (80/20) and 5-fold cross-validation. The primary metrics for evaluation are **F1-Score, ROC-AUC, and Recall (Sensitivity)**.

5. Key Risks and Assumptions

Key Risks and Mitigation:

Risk	Likelihood/Impact	Mitigation Strategy
Class Imbalance	High/Medium	Apply SMOTE, class weights, threshold optimization.
Overfitting	Medium/High	Use cross-validation, regularization, and pruning.
Model Interpretability	Medium/Medium	Use SHAP values and keep models explainable.

Assumptions:

- 1. The dataset adequately represents real-world telecom customer behavior.
- 2. Customer attributes (features) remain relevant predictors over time.
- 3. Churn labels are accurate and consistently defined.